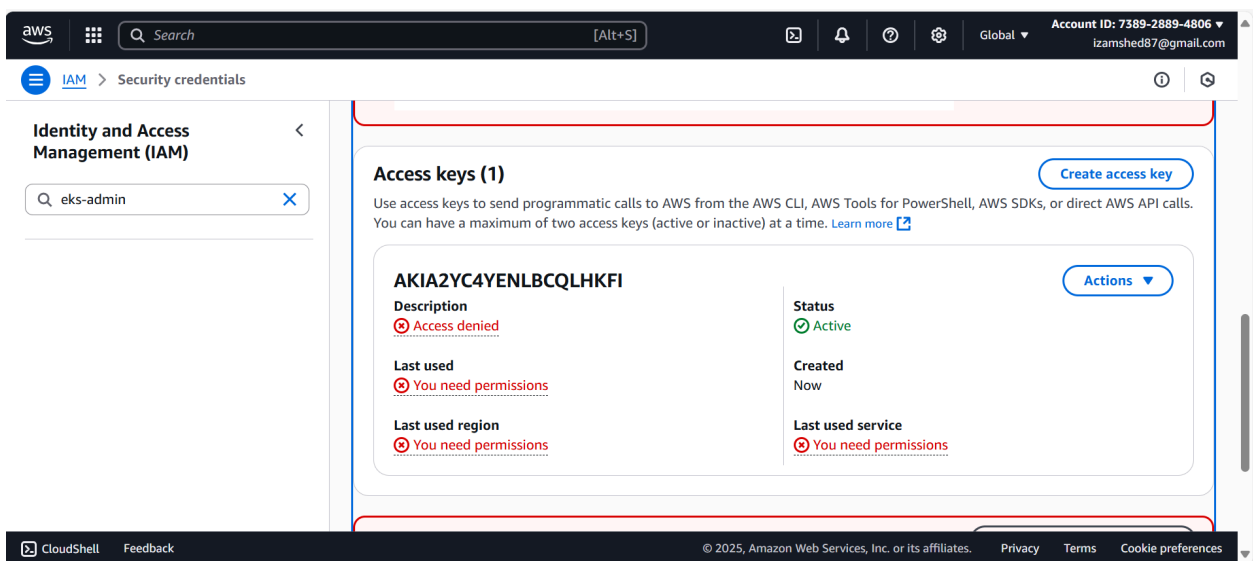


1.Project-Structure

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ ls
README.md  main.tf  outputs.tf  terraform.tfvars  variables.tf
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ pwd
/mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ |
```

2.Creating Access Key && Secret Key



3. Terraform initialization, validation

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform -v
Terraform v1.12.2
on linux_amd64
+ provider registry.terraform.io/hashicorp/aws v6.9.0
+ provider registry.terraform.io/hashicorp/random v3.7.2
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/random from the dependency lock file
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.9.0
- Using previously-installed hashicorp/random v3.7.2

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform validate
Success! The configuration is valid.
```

4. Terraform Execution Plan preview

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform plan
data.aws_ami.amazon_linux2: Reading...
data.aws_ami.amazon_linux2: Read complete after 3s [id=ami-0e2c86481225d3c51]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform will perform the following actions:

# aws_instance.web will be created
+ resource "aws_instance" "web" {
  + ami              = "ami-0e2c86481225d3c51"
  + arn              = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone = (known after apply)
  + disable_api_stop  = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized     = (known after apply)
  + enable_primary_ipv6 = (known after apply)
  + force_destroy     = false
  + get_password_data  = false
  + host_id           = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile = (known after apply)
  + id                = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle = (known after apply)
  + instance_state     = (known after apply)
  + instance_type      = "t2.micro"
  + ipv6_address_count = (known after apply)
  + ipv6_addresses     = (known after apply)
  + key_name           = (known after apply)
}
```

5.Terraform apply (Infra Creation)

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform apply --auto-approve
data.aws_ami.amazon_linux2: Reading...
data.aws_ami.amazon_linux2: Read complete after 3s [id=ami-0e2c86481225d3c51]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform will perform the following actions:

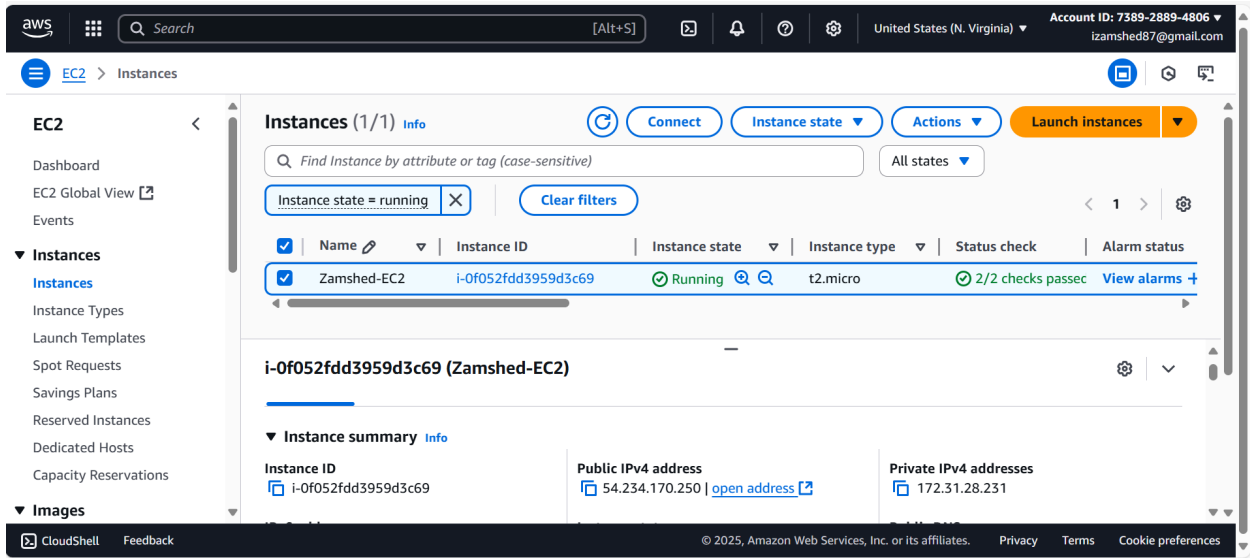
# aws_instance.web will be created
+ resource "aws_instance" "web" {
  + ami                  = "ami-0e2c86481225d3c51"
  + arn                  = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone     = (known after apply)
  + disable_api_stop     = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized        = (known after apply)
  + enable_primary_ipv6   = (known after apply)
  + force_destroy        = false
  + get_password_data     = false
  + host_id              = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile  = (known after apply)
  + id                   = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle   = (known after apply)
  + instance_state       = (known after apply)
  + instance_type        = "t2.micro"
  + ipv6_address_count    = (known after apply)
  + ipv6_addresses       = (known after apply)
  + key_name             = (known after apply)
}
```

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

```
aws_region = "us-east-1"
ec2_instance_id = "i-0f052fdd3959d3c69"
s3_bucket_name = "tf-assignment-9deb"
```

6.Output-EC2 & S3 in AWS

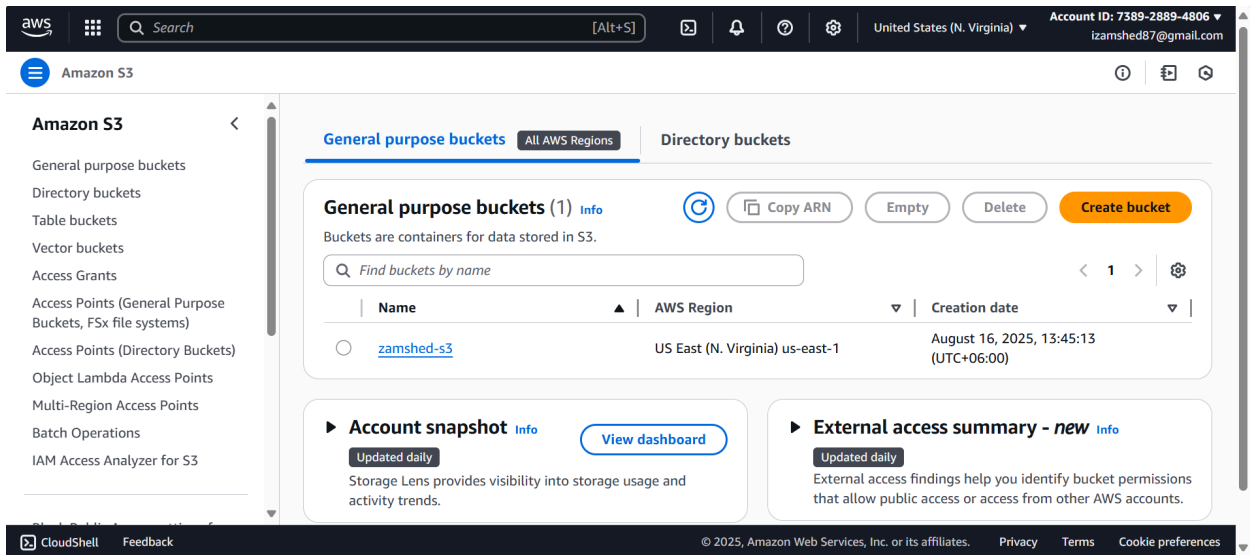


The screenshot displays the AWS Management Console for the EC2 service. The left sidebar shows the navigation menu with 'Instances' selected. The main content area shows a list of instances with one instance, 'Zamshed-EC2', in a 'Running' state. Below the list, the 'Instance summary' for 'i-0f052fdd3959d3c69' is shown, including its ID, public IPv4 address (54.234.170.250), and private IPv4 addresses (172.31.28.231).

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
Zamshed-EC2	i-0f052fdd3959d3c69	Running	t2.micro	2/2 checks passed	View alarms

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0f052fdd3959d3c69	54.234.170.250 open address	172.31.28.231



The screenshot displays the AWS Management Console for the Amazon S3 service. The left sidebar shows the navigation menu with 'Amazon S3' selected. The main content area shows the 'General purpose buckets' tab, which lists one bucket, 'zamshed-s3', in the 'US East (N. Virginia) us-east-1' region, created on August 16, 2025, at 13:45:13 (UTC+06:00).

Name	AWS Region	Creation date
zamshed-s3	US East (N. Virginia) us-east-1	August 16, 2025, 13:45:13 (UTC+06:00)

Account snapshot

Updated daily

Storage Lens provides visibility into storage usage and activity trends.

External access summary - new

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

7.Terraform Destroy (Infra Deletion)

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$ terraform destroy
data.aws_ami.amazon_linux2: Reading...
aws_s3_bucket.bucket: Refreshing state... [id=zamshed-s3]
data.aws_ami.amazon_linux2: Read complete after 4s [id=ami-0e2c86481225d3c51]
aws_instance.web: Refreshing state... [id=i-0f052fdd3959d3c69]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
  - destroy

Terraform will perform the following actions:

# aws_instance.web will be destroyed
- resource "aws_instance" "web" {
  - ami              = "ami-0e2c86481225d3c51" -> null
  - arn              = "arn:aws:ec2:us-east-1:738928894806:instance/i-0f052fdd3959d3c69" -> null
  - associate_public_ip_address = true -> null
  - availability_zone = "us-east-1b" -> null
  - disable_api_stop    = false -> null
  - disable_api_termination = false -> null
  - ebs_optimized        = false -> null
  - force_destroy        = false -> null
  - get_password_data     = false -> null
  - hibernation           = false -> null
  - id                  = "i-0f052fdd3959d3c69" -> null
  - instance_initiated_shutdown_behavior = "stop" -> null
  - instance_state       = "running" -> null
  - instance_type         = "t2.micro" -> null
  - ipv6_address_count    = 0 -> null
  - ipv6_addresses        = [] -> null
  - monitoring            = false -> null
  - placement_partition_number = 0 -> null
  - primary_network_interface_id = "eni-0a5c6e3905a14a763" -> null
}
```

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_s3_bucket.bucket: Destroying... [id=zamshed-s3]
aws_instance.web: Destroying... [id=i-0f052fdd3959d3c69]
aws_s3_bucket.bucket: Destruction complete after 2s
aws_instance.web: Still destroying... [id=i-0f052fdd3959d3c69, 00m10s elapsed]
aws_instance.web: Still destroying... [id=i-0f052fdd3959d3c69, 00m20s elapsed]
aws_instance.web: Destruction complete after 25s
```

Destroy complete! Resources: 2 destroyed.

```
zamshed@Zamshed-565177: /mnt/e/Downloads/Ostad/Assignments/Assignment-5/terraform-modules$
```